Lab 04 Report: Heart Disease Classification using Neural Network and AdaBoost

Course: ANN & Deep Learning · ·

1. Objective and Dataset

This lab builds and compares two binary classifiers that predict heart disease: a custom neural network (NN) and an AdaBoost ensemble. On a held- out test set of 61 patients, AdaBoost reached 90.2% accuracy and the neuralnetwork 88.5%, with the neural network achieving the higher recall (96.4%).

Item	Detail
Dataset	UCI Heart Disease, Cleveland processed version
Source	archive.ics.uci.edu/dataset/45/heart+disease
Samples	303 patients
Features	13 predictors + 1 target
Task	Binary classification: 0 = no disease, 1 = disease
Missing values	`ca` (4), `thal` (2), stored as `?` in the raw file

Feature descriptions

Feature	Type
age	Numerical
sex	Categorical (binary)
cp	Categorical
trestbps	Numerical
chol	Numerical
fbs	Categorical (binary)

Feature	Type
restecg	Categorical
thalach	Numerical
exang	Categorical (binary)
oldpeak	Numerical
slope	Categorical
ca	Numerical (ordinal count)
thal	Categorical

Descriptive statistics (`df.describe().T`)

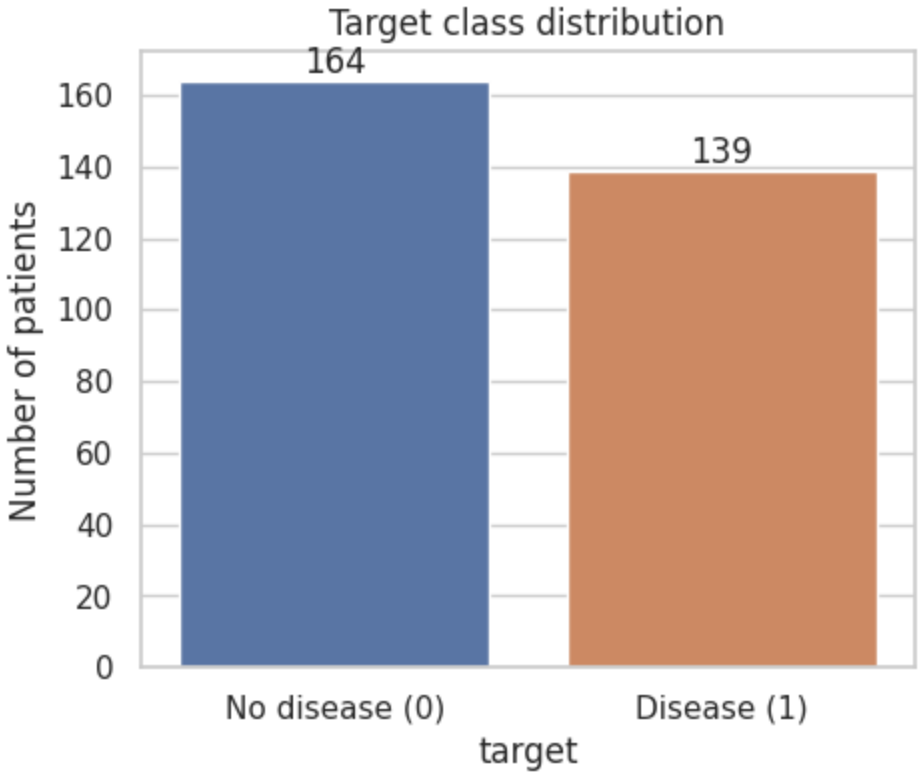
Feature	Count	Mean	Std	Min	25%	50%	75%	Max
age	303	54.44	9.04	29.0	48.0	56.0	61.0	77.0
sex	303	0.68	0.47	0.0	0.0	1.0	1.0	1.0
cp	303	3.16	0.96	1.0	3.0	3.0	4.0	4.0
trestbps	303	131.69	17.60	94.0	120.0	130.0	140.0	200.0
chol	303	246.69	51.78	126.0	211.0	241.0	275.0	564.0
fbs	303	0.15	0.36	0.0	0.0	0.0	0.0	1.0
restecg	303	0.99	0.99	0.0	0.0	1.0	2.0	2.0
thalach	303	149.61	22.88	71.0	133.5	153.0	166.0	202.0
exang	303	0.33	0.47	0.0	0.0	0.0	1.0	1.0
oldpeak	303	1.04	1.16	0.0	0.0	0.8	1.6	6.2
slope	303	1.60	0.62	1.0	1.0	2.0	2.0	3.0
ca	299	0.67	0.94	0.0	0.0	0.0	1.0	3.0
thal	301	4.73	1.94	3.0	3.0	3.0	7.0	7.0
target	303	0.46	0.50	0.0	0.0	0.0	1.0	1.0

Interpretation: mean age is 54.4 years (range 29–77); 68% of patients are male (mean of `sex`); mean cholesterol is 246.7 mg/dl with an outlier at 564. The counts of 299 for `ca` and 301 for `thal` confirm missing values.

4. Balanced vs. Imbalanced Data (Cell 5)

The dataset is approximately balanced: 164 patients (54.13%) have no disease and 139 (45.87%) have disease, a minority/majority ratio of 0.85.

Class	Count	Percentage
0 – No disease	164	54.13%
1 – Disease	139	45.87%



g Values and Data Quality (Cell 6)

Only 6 cells are missing (`ca` 4, `thal` 2) and there are no duplicate rows, so all 303 rows are kept and the gaps are imputed.

```
missing = df.isnull().sum()
df.duplicated().sum()   # 0
df.nunique()
```

Column	Dtype	Missing	Missing %	Unique values
age	float64	0	0.00	41
sex	float64	0	0.00	2
cp	float64	0	0.00	4
trestbps	float64	0	0.00	50
chol	float64	0	0.00	152
fbs	float64	0	0.00	2
restecg	float64	0	0.00	3
thalach	float64	0	0.00	91
exang	float64	0	0.00	2
oldpeak	float64	0	0.00	40
slope	float64	0	0.00	3
ca	float64	**4**	**1.32**	4
thal	float64	**2**	**0.66**	3
target	int64	0	0.00	2

Decision — impute rather than drop. Dropping rows would lose about 2% of an already small dataset. Instead:

- `ca` (an ordinal count) → **median** imputation, which is robust to outliers.
- `thal` (a category) → **most-frequent (mode)** imputation, since an average such as 4.73 has no meaning for a category.

Data Visualization

The strongest single signals for heart disease are a lower maximum heart rate, asymptomatic chest pain, more coloured vessels (`ca`) and higher ST depression (`oldpeak`).

6.1 Target distribution

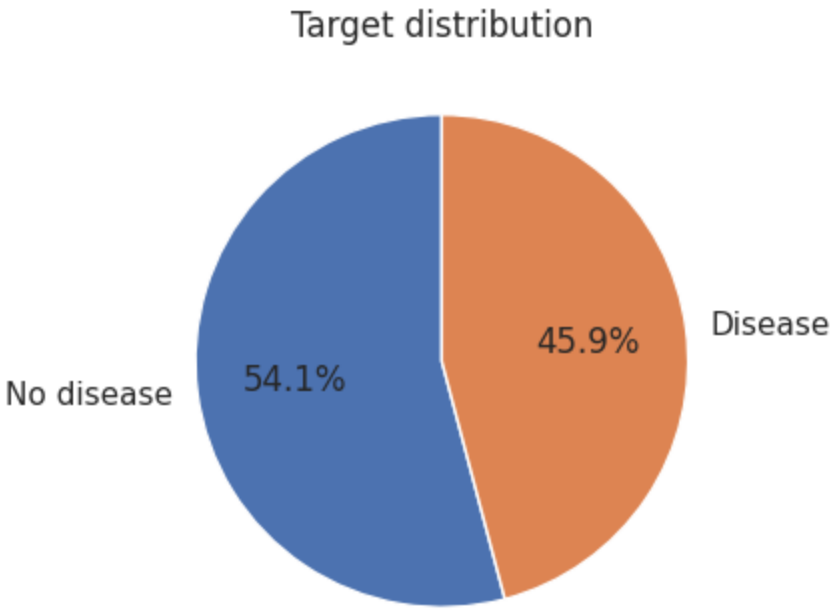


Figure 2. 54.1% no disease vs 45.9% disease — confirms the balance found in Cell 5.

6.2 Age distribution by target

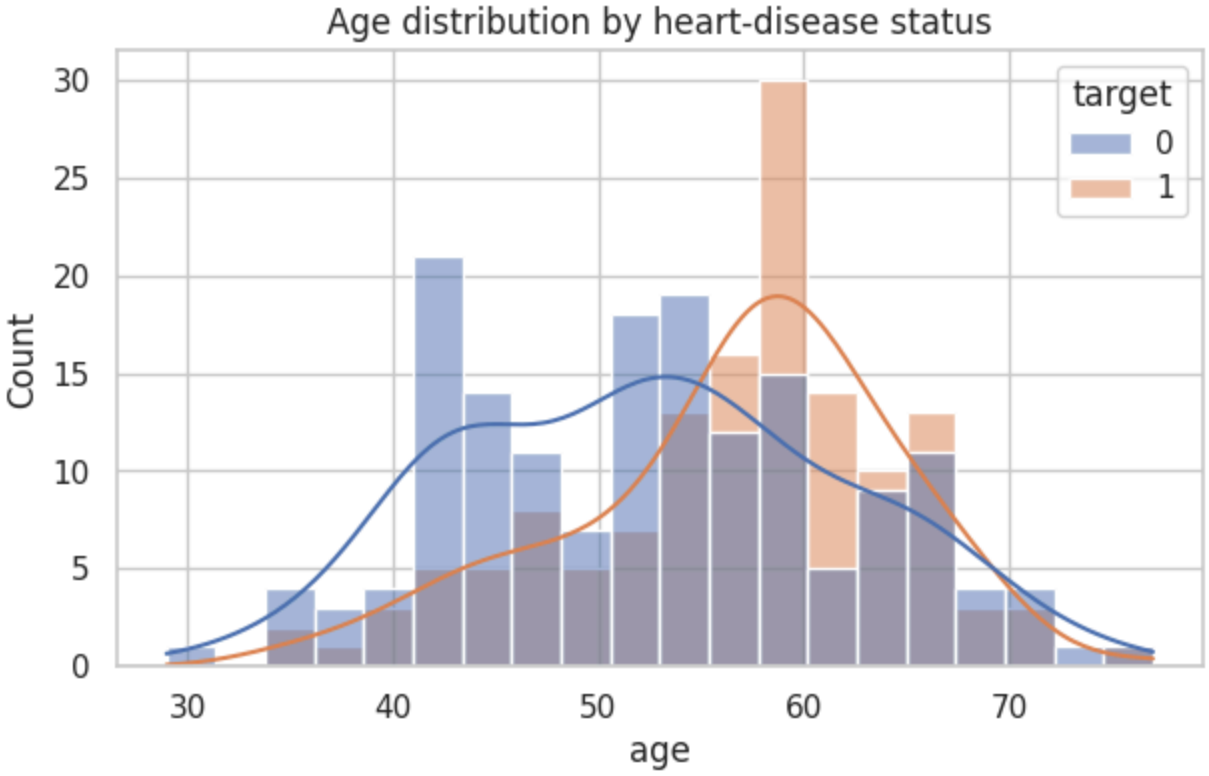


Figure 3. Histogram with KDE curves. Disease patients cluster in the 55–65 range, while the no-disease group is spread wider and includes more younger patients; risk rises with age.

6.3 Maximum heart rate (thalach) vs target

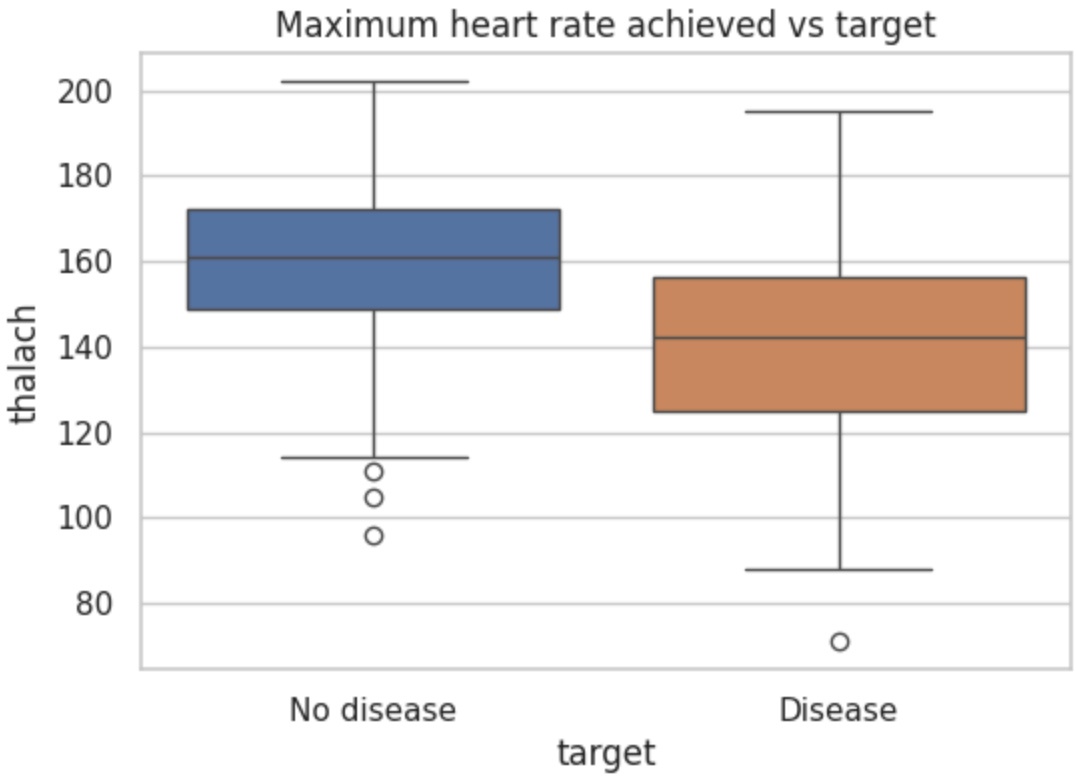


Figure 4. Patients with disease reach a clearly lower maximum heart rate (median ≈ 142 vs ≈ 161 bpm). In a boxplot the middle line is the median, the box holds the middle 50% of values and dots are outliers.

6.4 Chest pain type vs target

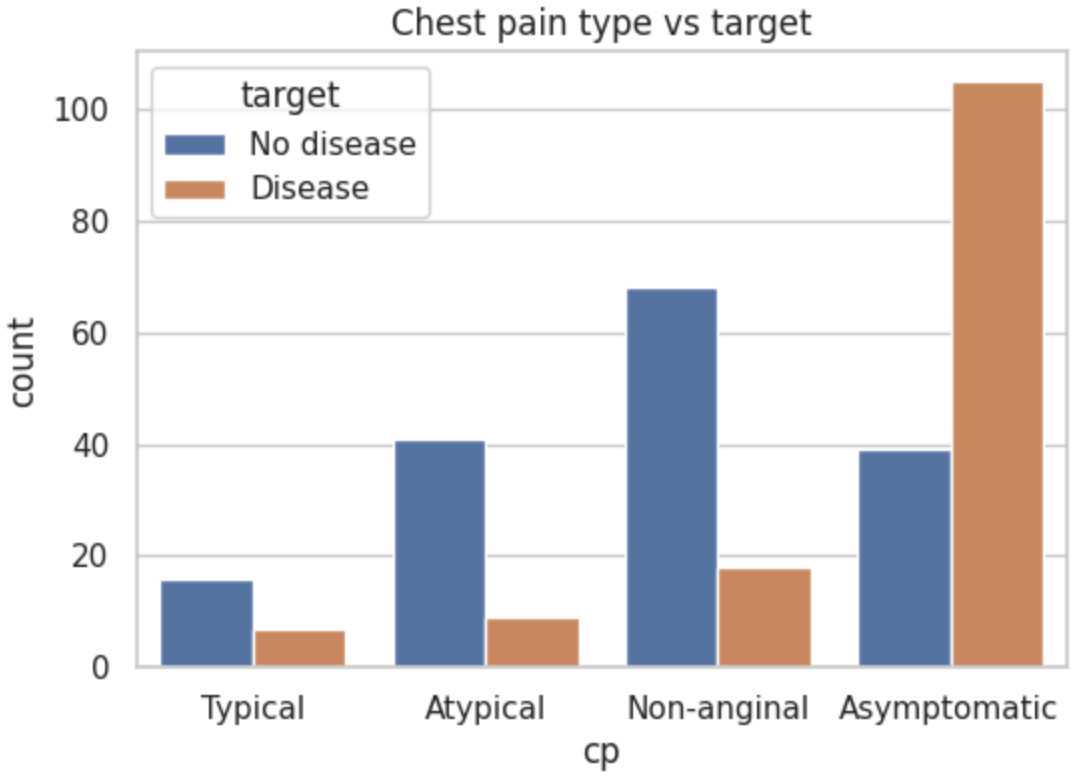


Figure 5. Most disease cases have **asymptomatic** chest pain (cp = 4), while typical, atypical and non-anginal pain are more common in healthy patients — a counter-intuitive but well-known pattern in this dataset.

6.5 Correlation heatmap

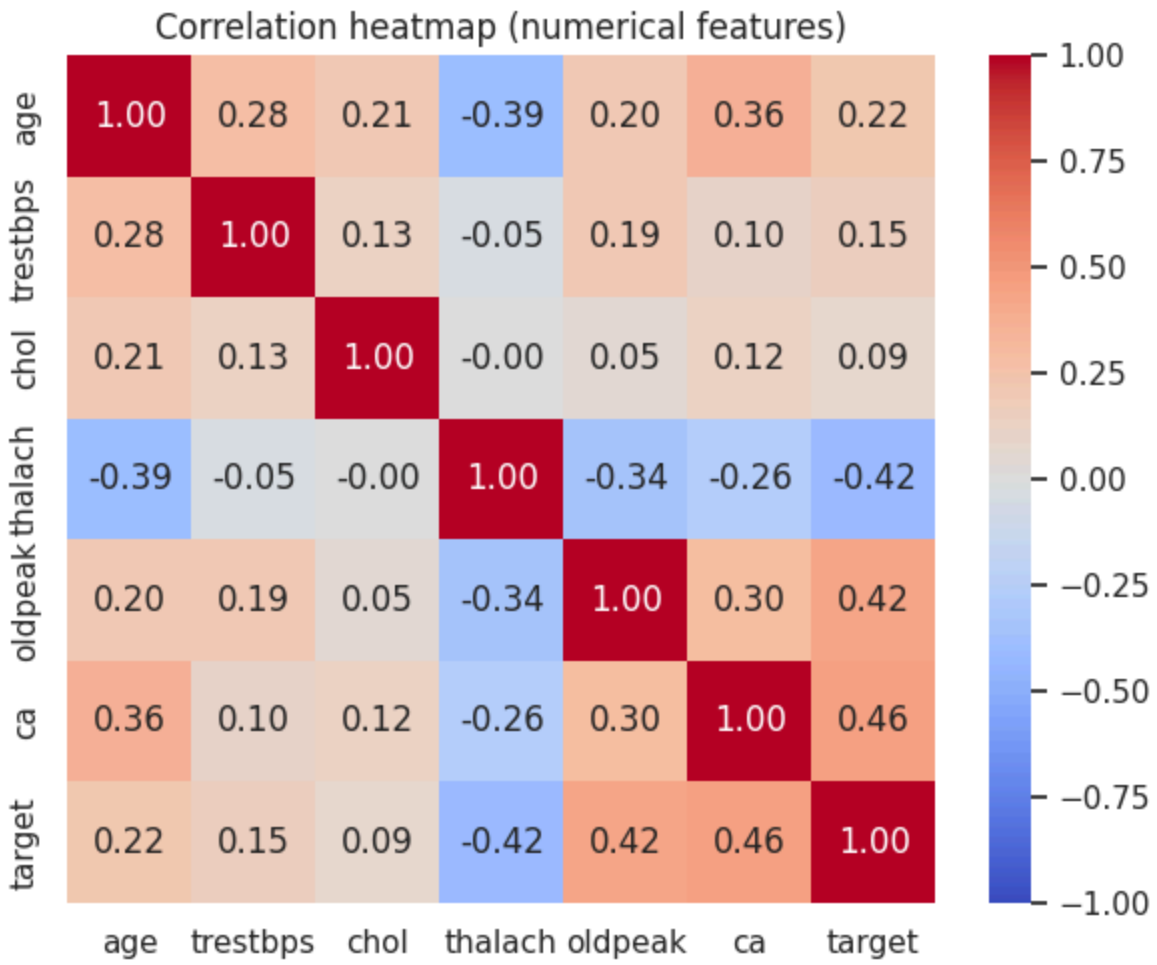


Figure 6. Pearson correlation runs from −1 (opposite movement) through 0 (no linear link) to +1 (move together).

Feature	Correlation with target
ca	+0.460
oldpeak	+0.425
age	+0.223
trestbps	+0.151
chol	+0.085
thalach	−0.417

`ca`, `oldpeak` and `thalach` are the most informative; `chol` and `trestbps` are weak. The largest correlation between predictors is about |0.4| (`age` – `thalach`), so there is no serious multicollinearity.

7. Preprocessing

After preprocessing, the 13 raw features become 22 model inputs, split into 205 training, 37 validation and 61 test samples, with all statistics learned from the training set only.

7.1 Steps

1. **Separate X and y:** `X = df.drop(columns=["target"])`, `y = df["target"]`.
2. **Feature groups:** numerical = `age, trestbps, chol, thalach, oldpeak, ca`; categorical = `sex, cp, fbs, restecg, exang, slope, thal`.
3. **Train/test split first:** 80/20, `stratify=y`, `random_state=42` → 242 train, 61 test.
4. **Numerical pipeline:** median imputation → `StandardScaler` (z = (x − mean) / std).
5. **Categorical pipeline:** most-frequent imputation → `OneHotEncoder(handle_unknown="ignore", drop="if_binary")`.
6. **ColumnTransformer:** applies each pipeline to its own columns.
7. **Fit on train only:** `fit_transform(X_train)` then `transform(X_test)`.
8. **Validation split:** 15% of the training set held out (stratified) → 205 train, 37 validation.
9. **Tensors + DataLoaders:** float32 tensors, targets reshaped to `[N, 1]`; batch size 32, shuffled for training.

Resulting feature count

Group	Columns
Numerical	6
sex	1
cp	4
fbs	1
restecg	3
exang	1

Group	Columns
slope	3
thal	3
Total	**22**

Output: `Train shape: (242, 22) Test shape: (61, 22)` and `NN train / val / test sizes: 205 37 61`.

8. Neural Network Design and Parameters (Cells 9–11)

The network is a 5-layer funnel (22 → 32 → 16 → 8 → 1) with 1,409 trainable parameters; the hand calculation matches the PyTorch count exactly.

8.1 Architecture (Cell 9)

Layer	Neurons	Activation
Input	22	None (features passed through)
Hidden 1	32	ReLU + Dropout 0.3
Hidden 2	16	ReLU + Dropout 0.3
Hidden 3	8	ReLU
Output	1	Sigmoid (inside `BCEWithLogitsLoss` / at prediction)

Design choices:

- **ReLU**, max(0, x), in hidden layers: simple, fast and avoids vanishing gradients.
- **Sigmoid output**, 1 / (1 + e^−x), with one neuron: turns the output into a probability of disease for a binary task.
- **Funnel shape (32 → 16 → 8):** compresses information step by step into the key patterns.
- **Small network + Dropout 0.3:** only 205 training samples, so a small model limits overfitting; dropout randomly disables 30% of neurons during training.

8.2 Manual parameter calculation (Cell 10)

```
\text{Parameters} = (\text{inputs} \times \text{outputs}) + \text{outputs (biases)}
```

Layer	Input units	Output units	Calculation	Parameters
Hidden 1	22	32	(22 × 32) + 32 = 704 + 32	736
Hidden 2	32	16	(32 × 16) + 16 = 512 + 16	528
Hidden 3	16	8	(16 × 8) + 8 = 128 + 8	136
Output	8	1	(8 × 1) + 1 = 8 + 1	9
Total			736 + 528 + 136 + 9	**1,409**

The input, ReLU and Dropout layers have no trainable parameters.

Layer	Shape	Params
Linear-0	22 → 32	736
Linear-3	32 → 16	528
Linear-6	16 → 8	136
Linear-8	8 → 1	9
Total trainable		**1,409**

Training output

Epoch	Train loss	Train acc	Val loss	Val acc
1	0.6890	0.541	0.6882	0.541
10	0.6331	0.541	0.6124	0.541
20	0.5380	0.698	0.4996	0.784
30	0.4224	0.834	0.3825	0.811
40	0.3512	0.878	0.3162	0.838

Epoch	Train loss	Train acc	Val loss	Val acc
50	0.3096	0.878	0.3079	0.865
60	0.3207	0.898	0.3225	0.865

`Early stopping at epoch 62` · `Best epoch: 42 (val loss 0.3043) -> weights restored`

output: the starting loss of 0.69 equals ln(2), the loss of random 50/50 guessing. The early accuracy of 0.541 equals the share of class 0, meaning the untrained model predicted "no disease" for everyone before it learned.

10. Training vs. Validation Loss (Cell 15)

There is no significant overfitting: both losses fell together from about 0.69 to 0.31, and validation loss stayed at or slightly below training loss.

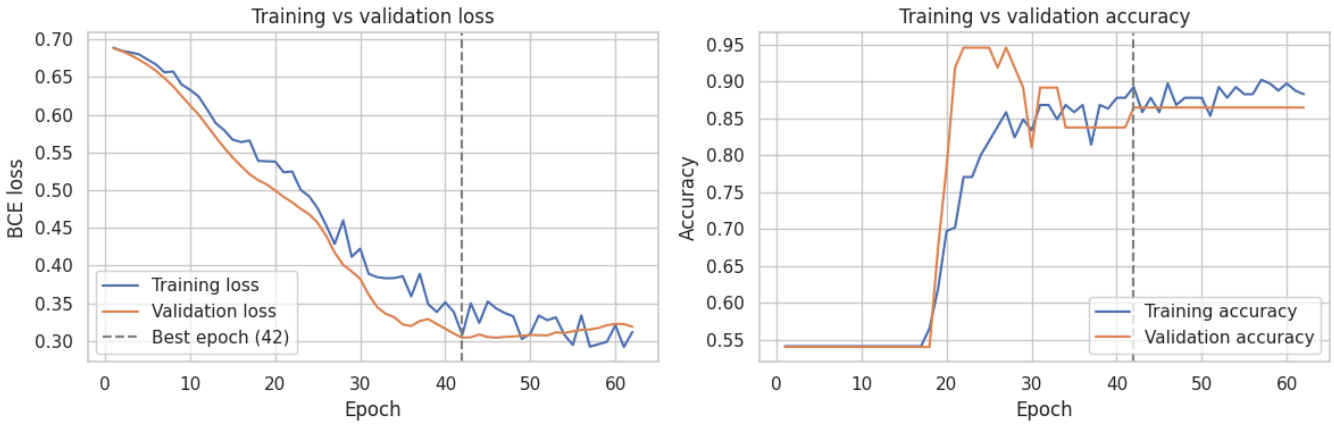


Figure 7. Left: training and validation BCE loss per epoch. Right: training and validation accuracy. The dashed line marks the best epoch (42).

Measure	Value
Final train loss	0.3119
Final validation loss	0.3189
Lowest validation loss	0.3043 at epoch 42

11. Neural Network Evaluation

	Predicted 0 (no disease)	Predicted 1 (disease)
Actual 0	27 (TN)	6 (FP)
Actual 1	1 (FN)	27 (TP)

- **TP** = disease, correctly detected · **TN** = healthy, correctly cleared
- **FP** = healthy but flagged (false alarm) · **FN** = disease missed (most dangerous in medicine)

11.2 Metrics

Metric	Formula	Calculation	Value
Accuracy	(TP + TN) / total	54 / 61	0.8852
Precision	TP / (TP + FP)	27 / 33	0.8182
Recall	TP / (TP + FN)	27 / 28	0.9643
F1-score	2·P·R / (P + R)	2 × 0.818 × 0.964 / 1.782	0.8852

12. AdaBoost Classifier

12.1 How AdaBoost works

1. Train a **weak learner** — a decision stump (`max_depth=1`) that asks a single yes/no question.
2. Find the samples it misclassified.
3. Increase the weights of those samples so the next stump focuses on them.
4. Repeat for `n_estimators = 100` rounds.
5. Predict with a **weighted vote** of all stumps; more accurate stumps get a larger say.

AdaBoost is a **boosting** ensemble: models are built sequentially, each correcting the last. This differs from **bagging** (e.g. Random Forest), which builds models in parallel and averages them. `learning_rate = 0.5` shrinks each stump's contribution for more stable learning. AdaBoost trained on all 242 training samples, as it needs no validation set.

12.2 Results

Metric	Value
Accuracy	0.9016
Precision	0.8667
Recall	0.9286
F1-score	0.8966
Training accuracy	0.876

	Predicted 0	Predicted 1
Actual 0	29 (TN)	4 (FP)
Actual 1	2 (FN)	26 (TP)

Training accuracy (0.876) is close to test accuracy (0.902), so AdaBoost is not overfitting.

12.3 Top feature importances

Rank	Feature	Importance
1	thalach	0.152
2	chol	0.132
3	ca	0.118
4	trestbps	0.098
5	oldpeak	0.097
6	age	0.080
7	cp = 4 (asymptomatic)	0.073
8	sex = 1 (male)	0.071

The top features largely agree with the correlation heatmap (`thalach` , `ca` , `oldpeak`) and the chest-pain plot (`cp = 4`).

13. Model Comparison (Cell 19)

AdaBoost is marginally better on accuracy, precision and F1, while the neural network has the higher recall; the gap is only 1–2 test patients, so the models perform comparably.

Model	Accuracy	Precision	Recall	F1-score
Neural Network	0.8852	0.8182	**0.9643**	0.8852
AdaBoost	**0.9016**	**0.8667**	0.9286	**0.8966**

